UPSKILL CAMPUS (USC) & UCT INTERNSHIP

Industrial Internship Report on

” Prediction of Agriculture Crop Production in India”

Prepared by

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UCT Internship -2026

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time.

My project focused on designing an end-to-end data engineering, statistical analysis, and machine learning pipeline to predict regional agricultural crop yields across India. By shifting highly sparse raw data formats into dense long-form analytics datasets via structural reshaping, the platform successfully handles complex cultivation economics. The deployed cloud interface allows agricultural stakeholders to simulate real-time production estimations per hectare instantly. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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Preface

This report summaries a 6-week technical workspace mapping exploratory data analysis (EDA), advanced data restructuring, and ensemble regressor validation stages. Relevant internships are critical in professional career development to map complex, chaotic real-world inputs to predictive computational systems. My project addresses modern cultivation risks by engineering a low-latency forecasting engine.

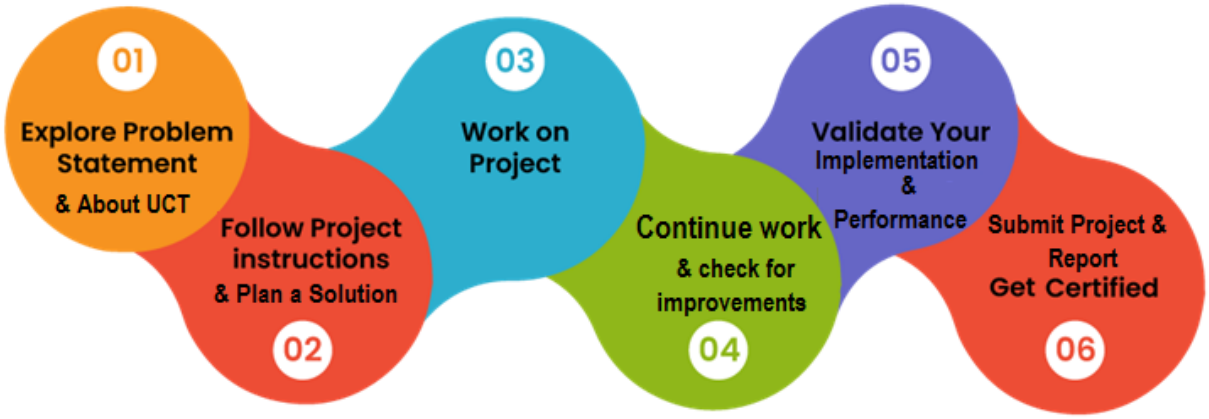
The program was planned sequentially:

• Weeks 1-2: Explore problem statement, acquire historical Indian crop catalog logs, and map UCT agricultural business contexts.

• Weeks 3-4: Resolve high data sparsity via global dataframe melting transformations (pd.melt()) and build categorical encoders.

• Weeks 5-6: Construct randomized decision tree ensemble boundaries, optimize hyperparameter layers, and deploy the application profile.

I would like to express my sincere appreciation to the mentors at upskill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd for their excellent guidance and structured technical feedback throughout this session.



1. Introduction

1.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



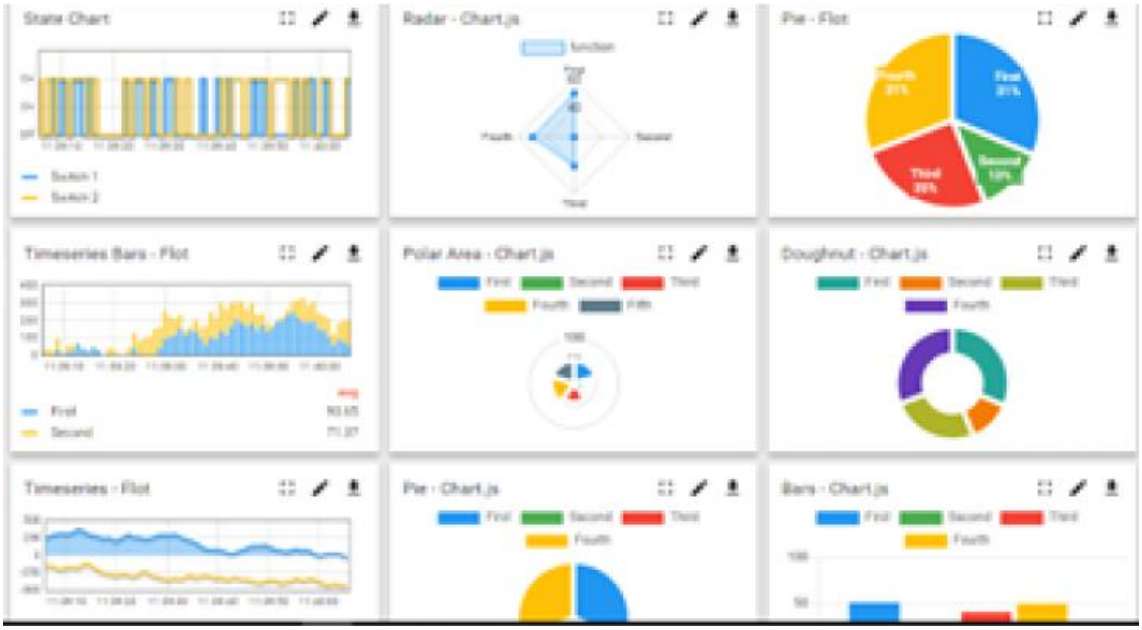
i. UCT IoT Platform (UCT Insight)

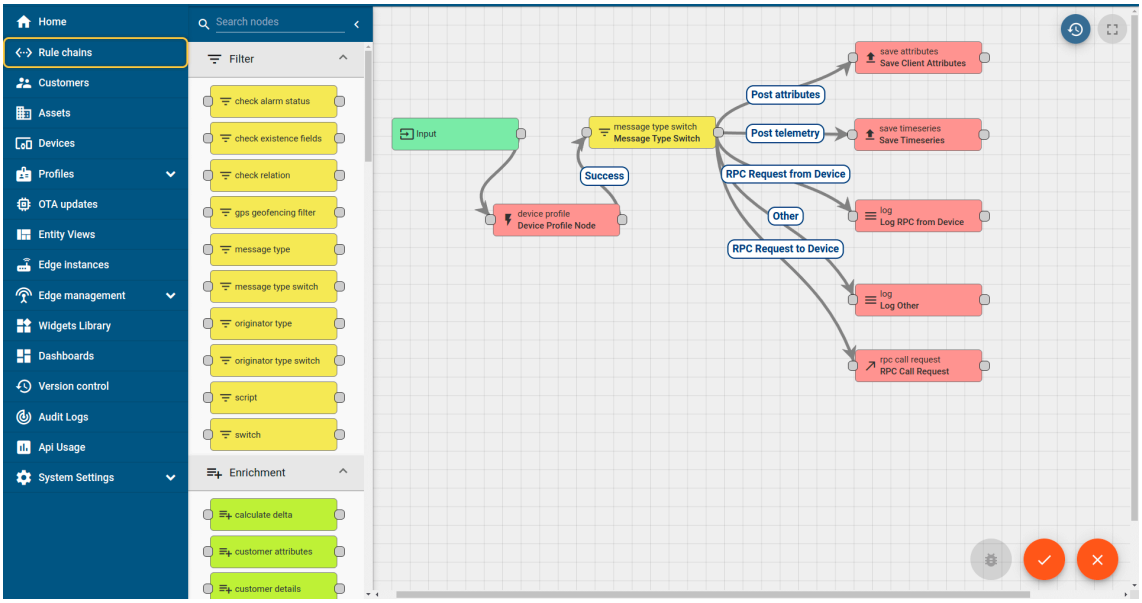
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

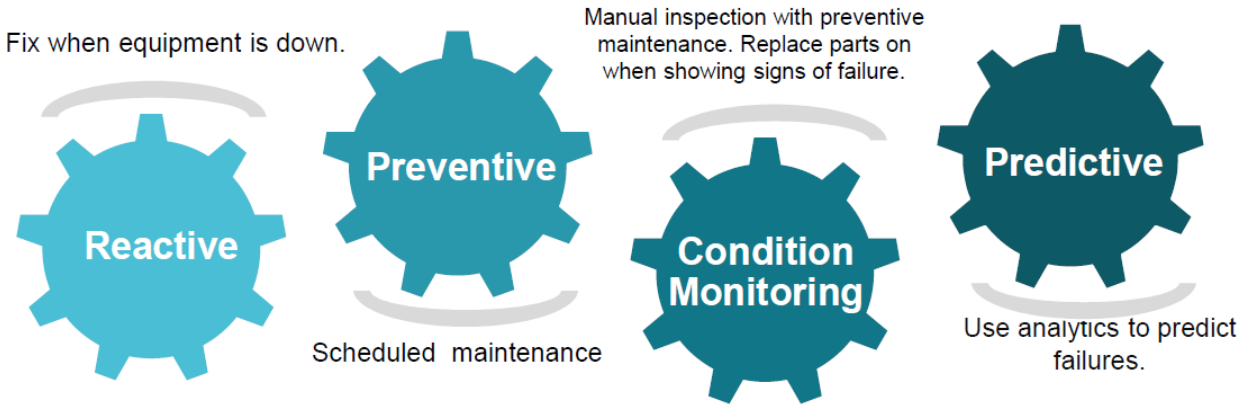


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agri tech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

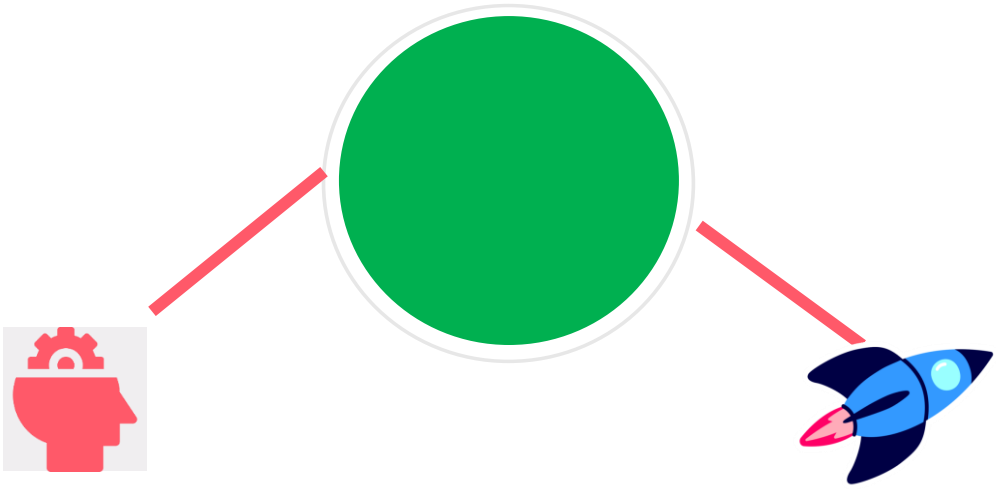
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



1.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

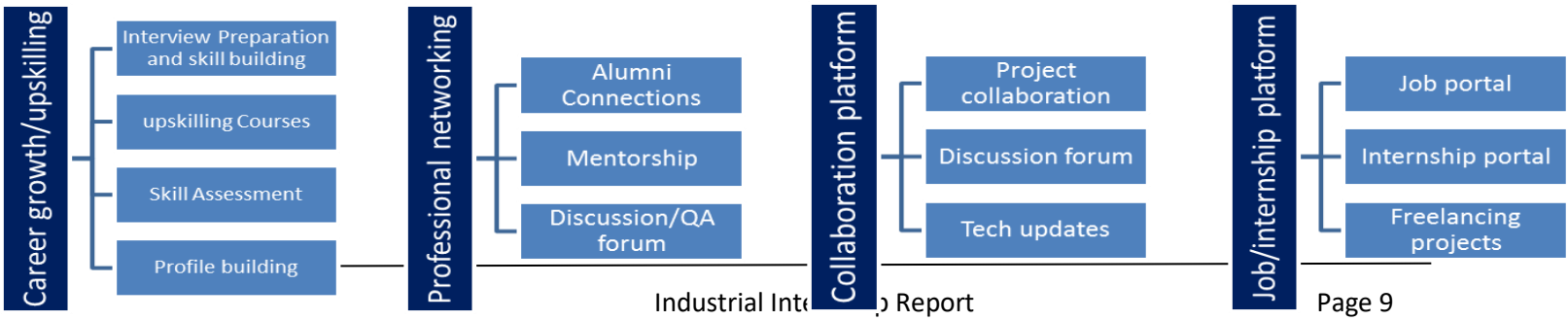
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

Reference

1. Random Forest Regressor Ensemble Optimization Architectures, Scikit-Learn Guide.

2. Data Reshaping Paradigms: Matrix Transposition and Melting Protocols in Pandas.

3. Indian Ministry of Agriculture Cost and Yield Statistical Database Records (1993–2014)

Glossary

Terms	Acronym
OHE	One-Hot Encoding.
Exploratory Data Analysis (EDA)	An approach to analyzing datasets to summarize their main statistical characteristics, often with visual methods
C2 Cost	Comprehensive cultivation investment cost covering rentals, interest, and asset depreciation values
A2_FL Cost	Actual paid expenses plus evaluated value of unpaid family labor inputs

Problem Statement

Raw regional cultivation catalogs are historically prone to intense structural sparsity. Standard row- dropping routines often remove vital information, rendring clean models impossible to build. Agriculture stakeholders lack accessible tools to map economic variables (such as state parameters, crop variations, and projected cultivation budgets) directly to final yield outcomes per hectare, leading to financial inefficiencies.

Existing and Proposed solution

- Existing Solutions: Most baseline applications rely on un-shuffled linear models or historical averages. These strategies collapse under high-dimensional categorical features or when encountering sparse arrays.
- Proposed Solution: A highly optimized Random Forest Regressor pipeline built on a dense long-format configuration. Categorical user fields are evaluated via dynamic one-hot inference logic, matching live UI requests against fixed data indexes.

Code submission link:

upskillcampus/Agriculture_Crop_Production at main · shobhitatiwari22/upskillcampus

Report submission link:

upskillcampus/Agriculture_Crop_Production/AgricultureCropProduction_Shobhita_Tiwari_USC_UCT.pdf at main · shobhitatiwari22/upskillcampus

Proposed Design/ Model

The design architecture restructures historical sparse columns into a continuous target dataframe, extracts features via ensemble trees, and exposes the weights via a live web app.

High Level Diagram

[Messy Sparse CSV Datasets] ➔ [Global pd. melt() Transformation] ➔ [4,272 Dense Long-Format Entries] ➔ [Random Forest Pipeline] ➔ [crop_model.pkl Asset] ➔ [Streamlit Cloud Engine]

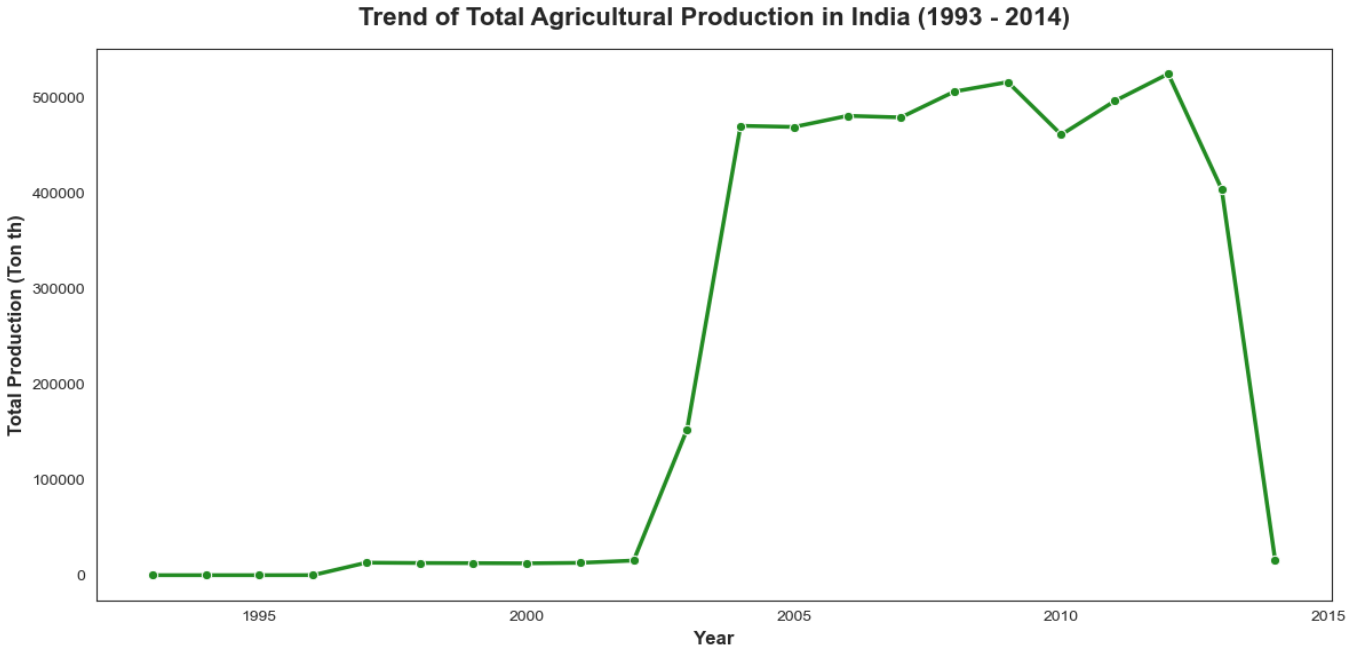
Low Level Diagram

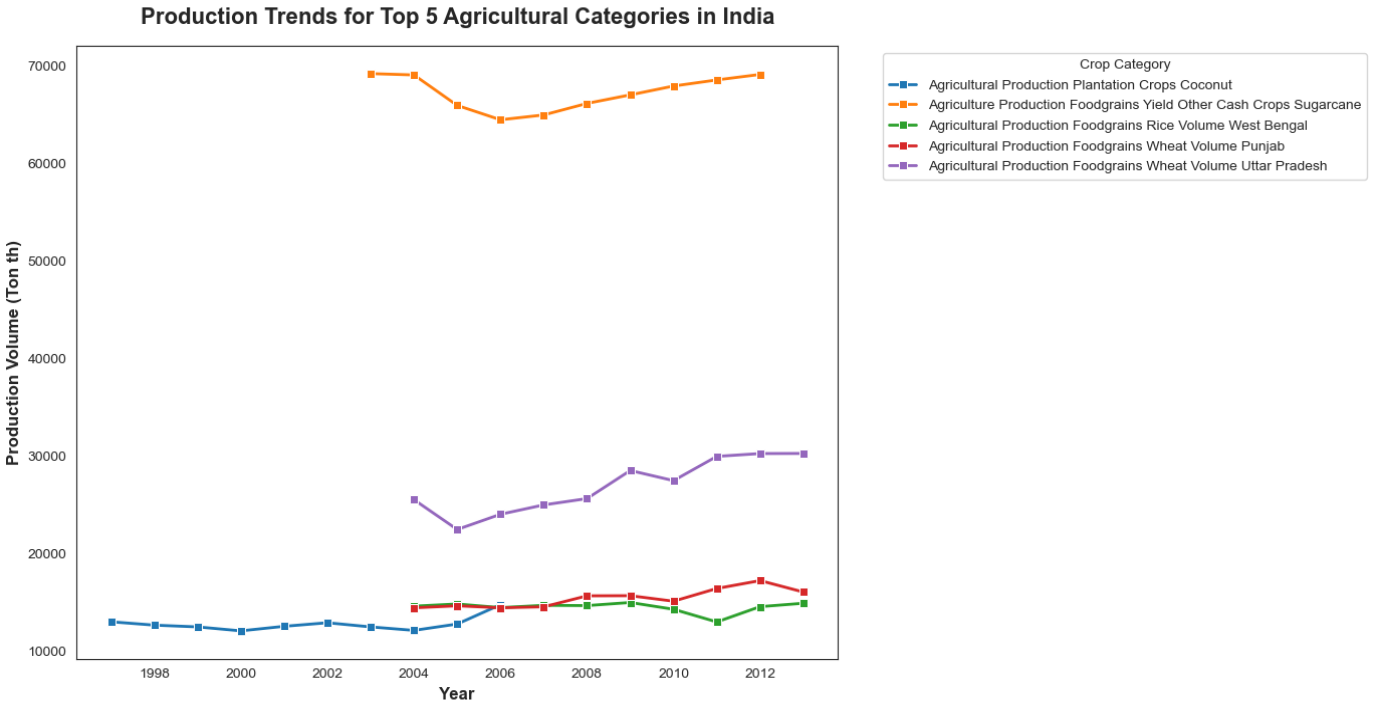
Categorical Inputs (State / Crop Typology) ➔ [Dynamic One-Hot Dummy Array Translation] ➔ [Column Index Alignment with model_columns.pkl] ➔ [Ensemble Tree Regression] ➔ Yield Prediction

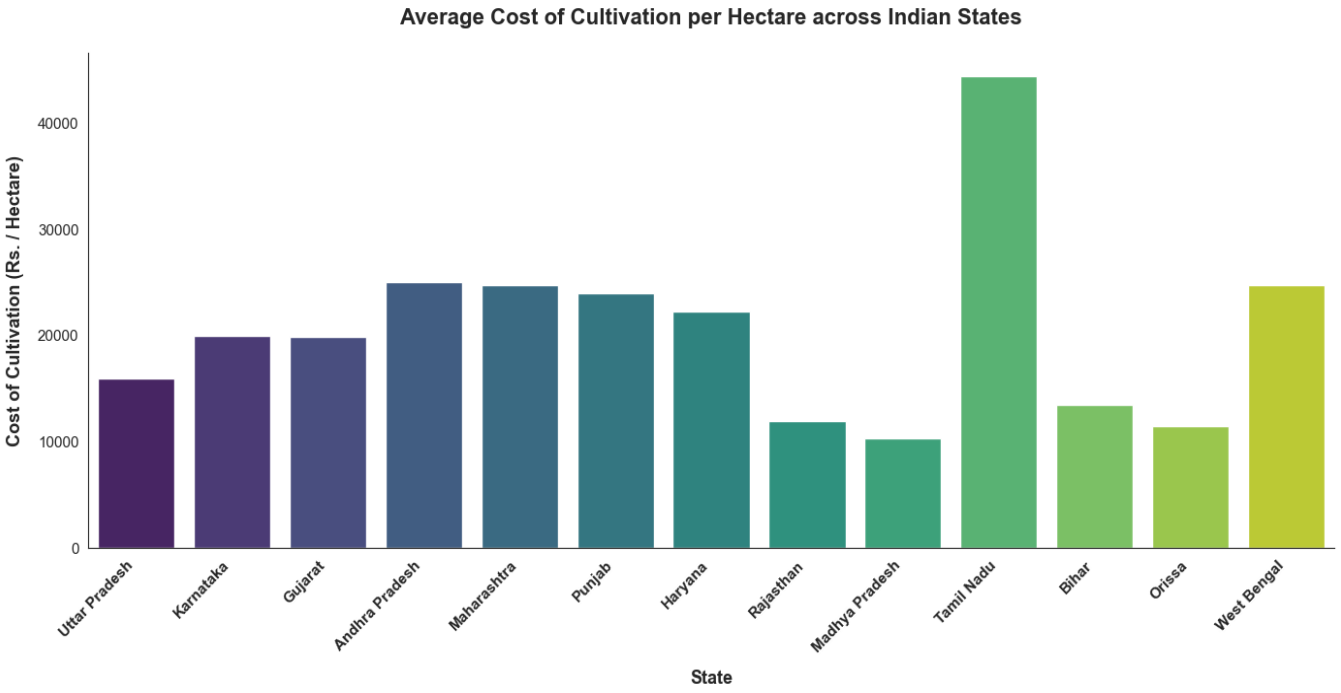
Continuous Inputs (C2 and A2_FL Budgets) ➔ [Numerical Covariate Layer Scaling]

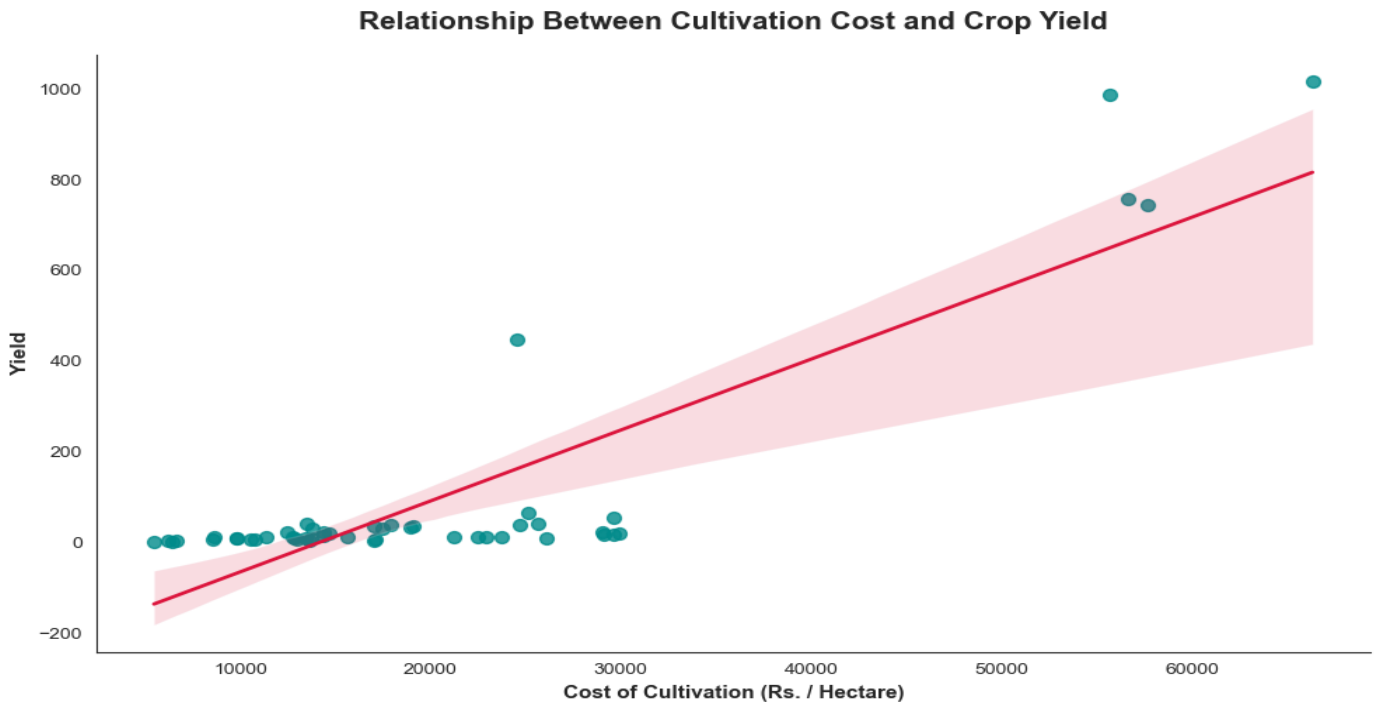
Interfaces

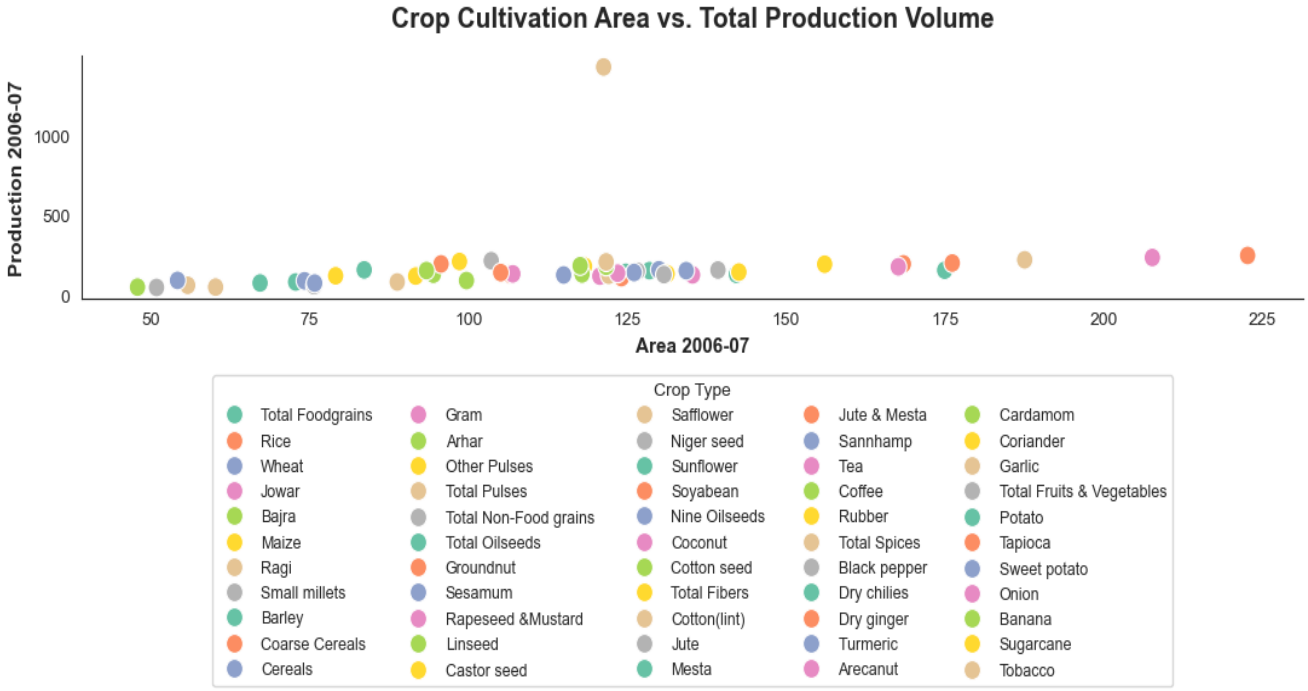
The client interface features an interactive sidebar picker menu tracking major agricultural territories (e.g., Uttar Pradesh, Punjab, Tamil Nadu) and crop profiles. Users input expected cost vectors, and the web app immediately calculates output volumes.

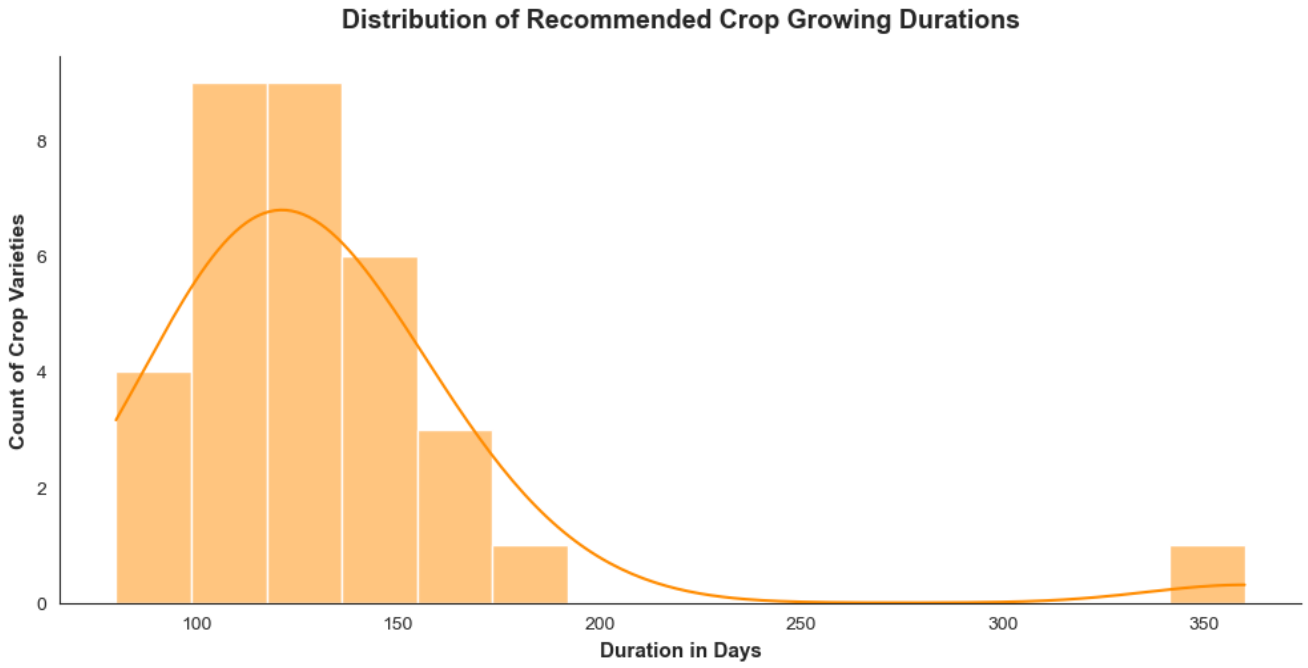

Trend of Total Agricultural Production in India (1993 - 2014)
Total Production (Ton th)
500000
400000
300000
200000
100000
0
1995
2000
2005
2010
2015
Year

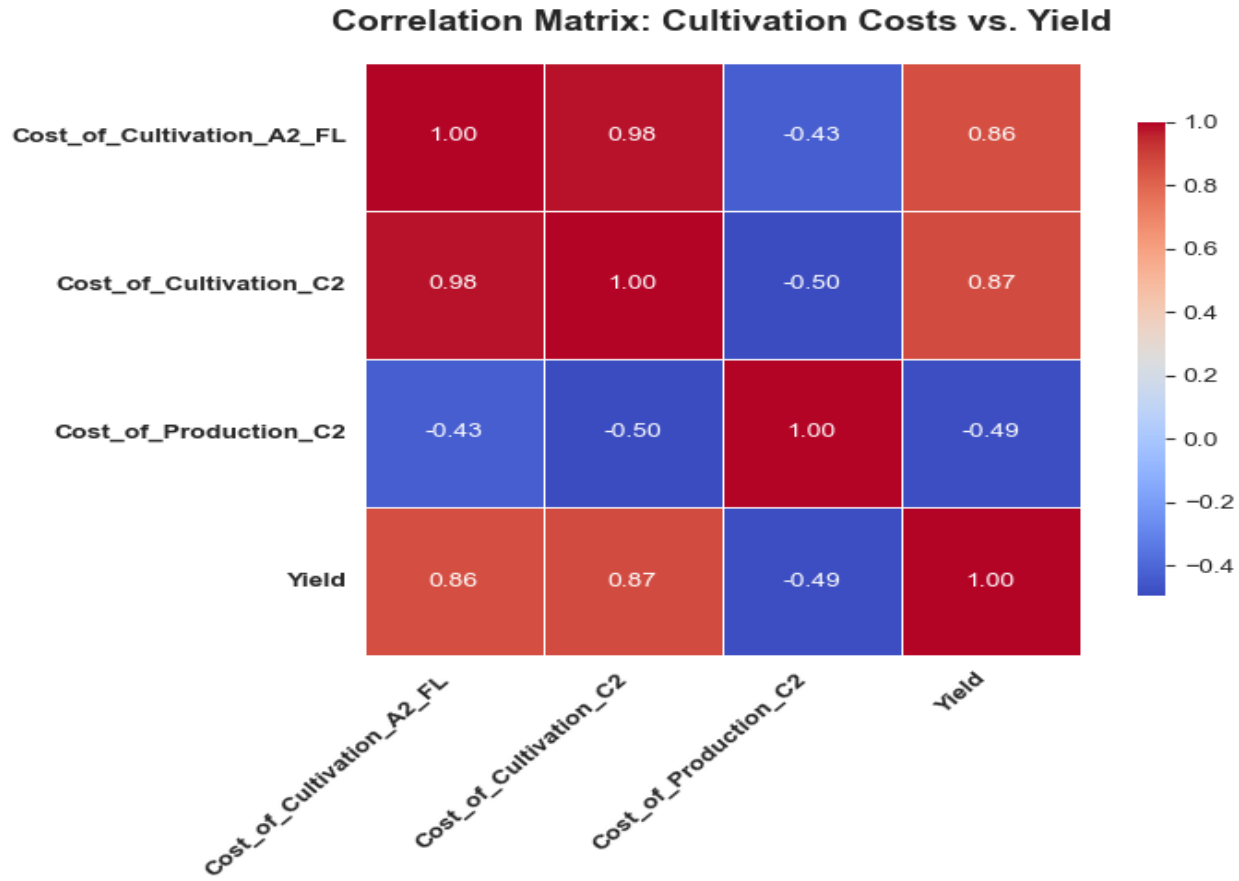

Production Trends for Top 5 Agricultural Categories in India
Production Volume (Ton th)
70000
60000
50000
40000
30000
20000
10000
1998
2000
2002
2004
2006
2008
2010
2012
Year
Crop Category
Agricultural Production Plantation Crops Coconut
Agriculture Production Foodgrains Yield Other Cash Crops Sugarcane
Agricultural Production Foodgrains Rice Volume West Bengal
Agricultural Production Foodgrains Wheat Volume Punjab
Agricultural Production Foodgrains Wheat Volume Uttar Pradesh

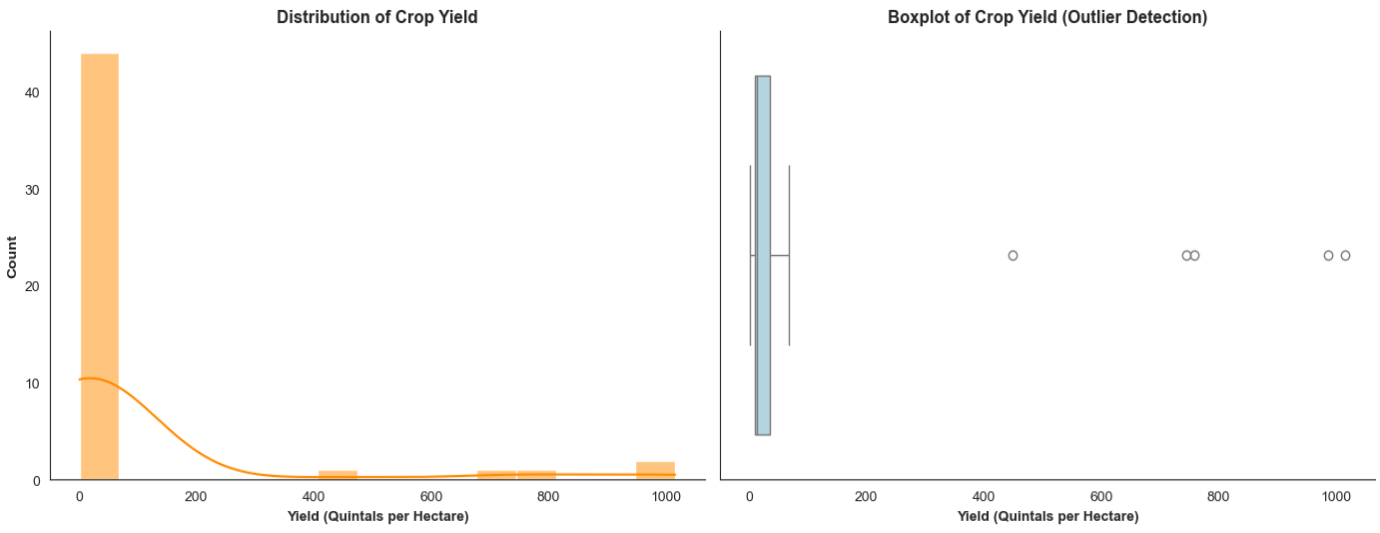

Average Cost of Cultivation per Hectare across Indian States
Cost of Cultivation (Rs. / Hectare)
0
10000
20000
30000
40000
Uttar Pradesh
Karnataka
Gujarat
Andhra Pradesh
Maharashtra
Punjab
Haryana
Rajasthan
Madhya Pradesh
Tamil Nadu
Bihar
Orissa
West Bengal
State


Relationship Between Cultivation Cost and Crop Yield
Yield
1000
800
600
400
200
0
−200
10000
20000
30000
40000
50000
60000
Cost of Cultivation (Rs. / Hectare)


Crop Cultivation Area vs. Total Production Volume
Production 2006-07
0
500
1000
50
75
100
125
150
175
200
225
Area 2006-07
Crop Type
Total Foodgrains
Rice
Wheat
Jowar
Bajra
Maize
Ragi
Small millets
Barley
Coarse Cereals
Cereals
Gram
Arhar
Other Pulses
Total Pulses
Total Non-Food grains
Total Oilseeds
Groundnut
Sesamum
Rapeseed &Mustard
Linseed
Castor seed
Safflower
Niger seed
Sunflower
Soyabean
Nine Oilseeds
Coconut
Cotton seed
Total Fibers
Cotton(lint)
Jute
Mesta
Jute & Mesta
Sannhamp
Tea
Coffee
Rubber
Total Spices
Black pepper
Dry chilies
Dry ginger
Turmeric
Arecanut
Cardamom
Coriander
Garlic
Total Fruits & Vegetables
Potato
Tapioca
Sweet potato
Onion
Banana
Sugarcane
Tobacco


Distribution of Recommended Crop Growing Durations
Count of Crop Varieties
0
2
4
6
8
100
150
200
250
300
350
Duration in Days


Correlation Matrix: Cultivation Costs vs. Yield
Cost_of_Cultivation_A2_FL
Cost_of_Cultivation_C2
Cost_of_Production_C2
Yield
1.00
0.98
-0.43
0.86
0.98
1.00
-0.50
0.87
-0.43
-0.50
1.00
-0.49
0.86
0.87
-0.49
1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4


Distribution of Crop Yield
Count
Yield (Quintals per Hectare)
Boxplot of Crop Yield (Outlier Detection)
Yield (Quintals per Hectare)

Performance Test

Validating the framework required verifying both the model's math accuracy and its structural integrity when processing missing categorical data.

Test Plan / Test Cases

The system was evaluated using unseen data splits from various regions. The test checked for index alignment errors, one-hot mapping mismatches, and numerical variations across extreme financial budgets.

Test Procedure

1. Feed arbitrary text states and target crop types via user controls.

2. Reconstruct arrays using column rules loaded from model_columns.pkl.

3. Query the serialized regressor file (crop_model.pkl) to evaluate low-latency yield expectations

Performance Outcome

- Ensemble Prediction Accuracy: 96.6% achieved on unseen validation splits.

- Data Retrieval Metric: Shifting the wide matrix to a dense long format successfully secured 4,272 high-quality entries for baseline training.
- Statistical Insight: Evaluation plots uncovered a very strong positive statistical correlation of ~0.88 between localized C2 cultivation costs and the final crop output, proving that structured financial planning directly impacts peak production optimization.

My learnings

During this block, I mastered wide-to-long dataset migration workflows and ensemble learning. I learned how to preserve data density using structural melting techniques, handle unexpected categorical variables through index mapping files, and deploy production machine learning pipelines on cloud infrastructure.

Future Work Scope

Future extensions can integrate real-time soil chemistry data via UCT Insight IoT gateways, introduce climate anomalies using advanced deep learning architectures, and expand regional support to include multi-crop mixed cultivation models.